Q1. JEE Main 2026 (23 January Shift 1)

In a perfectly inelastic collision, two spheres made of the same material with masses 15 kg and 25 kg, moving in opposite directions with speeds of $10 \mathrm{~m/s}$ and $30 \mathrm{~m/s}$, respectively, strike each other and stick together. The rise in temperature (in $^{\circ}\mathrm{C}$), if all the heat produced during the collision is retained by these spheres, is : (specific heat of sphere material $31 \mathrm{cal/kg.}\ ^{\circ}\mathrm{C}$ and $1 \mathrm{cal} = 4.2 \mathrm{~J}$)

(1) 1.15 (2) 1.75 (3) 1.44 (4) 1.95

Q2. JEE Main 2026 (22 January Shift 2)

Given below are two statements :

Statement I: For a mechanical system of many particles total kinetic energy is the sum of kinetic energies of all the particles.

Statement II: The total kinetic energy can be the sum of kinetic energy of the center of mass w.r.t to the origin and the kinetic energy of all the particles w.r.t. the center of mass as the reference.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are false

(3) Statement I is false but Statement II is true

(4) Both Statement I and Statement II are true

Q3. JEE Main 2026 (23 January Shift 2)

A body of mass 14 kg initially at rest explodes and breaks into three fragments of masses in the ratio 2 : 2 : 3. The two pieces of equal masses fly off perpendicular to each other with a speed of $18 \mathrm{~m/s}$ each. The velocity of the heavier fragment is ____ $\mathrm{m/s}$.

(1) 12 (2) $10\sqrt{2}$ (3) $24\sqrt{2}$ (4) $12\sqrt{2}$

ANSWERS AND SOLUTIONS

1. (3) **2.** (4) **3.** (4)

1. **(3)** Using conservation of momentum:

$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$ where $v_2 = -30$ m/s (opposite direction).

$15(10) + 25(-30) = 40 v_f \Rightarrow v_f = -15$ m/s

Initial kinetic energy:

$KE_i = \frac{1}{2}(15)(10)^2 + \frac{1}{2}(25)(30)^2 = 750 + 11250 = 12000$ J

Final kinetic energy:

$KE_f = \frac{1}{2}(40)(15)^2 = 4500$ J

Heat produced:

$Q = 12000 - 4500 = 7500 \text{ J} = \frac{7500}{4.2} = 1785.71$ cal

Temperature rise:

$\Delta T = \frac{Q}{m_{total} \cdot c} = \frac{1785.71}{40 \times 31} = 1.44^{\circ} C$

2. **(4)** Statement I: Total kinetic energy of a system equals $\sum_i \frac{1}{2} m_i v_i^2$. This is TRUE by definition.

Statement II: Total KE can be decomposed as $KE_{total} = \frac{1}{2} M v_{cm}^2 + \sum_i \frac{1}{2} m_i v_i'^2$, where the first term is kinetic energy of center of mass motion and the second is kinetic energy of particles relative to the center of mass. This is TRUE by the energy decomposition theorem (Koenig's theorem).

Both statements are true.

3. **(4)** From mass ratio 2:2:3 with total mass 14 kg: individual masses are 4 kg, 4 kg, and 6 kg.

Setting coordinate axes along the motion directions of the two equal fragments: fragment 1 has momentum $\vec{p_1} = 4 \times 18\hat{i} = 72\hat{i}$ kg·m/s, and fragment 2 has $\vec{p_2} = 4 \times 18\hat{j} = 72\hat{j}$ kg·m/s.

By momentum conservation, initially at rest: $\vec{p_1} + \vec{p_2} + \vec{p_3} = 0$, so $\vec{p_3} = -72\hat{i} - 72\hat{j}$ kg·m/s.

Velocity of 6 kg fragment: $\vec{v_3} = \frac{\vec{p_3}}{6} = -12\hat{i} - 12\hat{j}$ m/s.

Speed: $|\vec{v_3}| = \sqrt{144 + 144} = \sqrt{288} = 12\sqrt{2}$ m/s.